

1. Title: AI Powered Meeting Summarizer & Action Item Generator .

2. Domain: Artificial Intelligence / Natural Language Processing (NLP)

3. Application

- Develop an AI-based application that processes meeting transcripts or text discussions.
- Automatically generate clear and concise summaries of meetings.
- Extract key action items and responsibilities from the discussion.
- Improve productivity for students, professionals, and organizations by reducing manual note-taking and saving time.

4. Hardware & Software Required

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: 4 GB minimum
- Storage: 100 GB or more
- Internet connection

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Python
- AI/NLP Libraries: TensorFlow or PyTorch
- Tool/IDE: VS Code or Jupyter Notebook

5. Input Required for Project

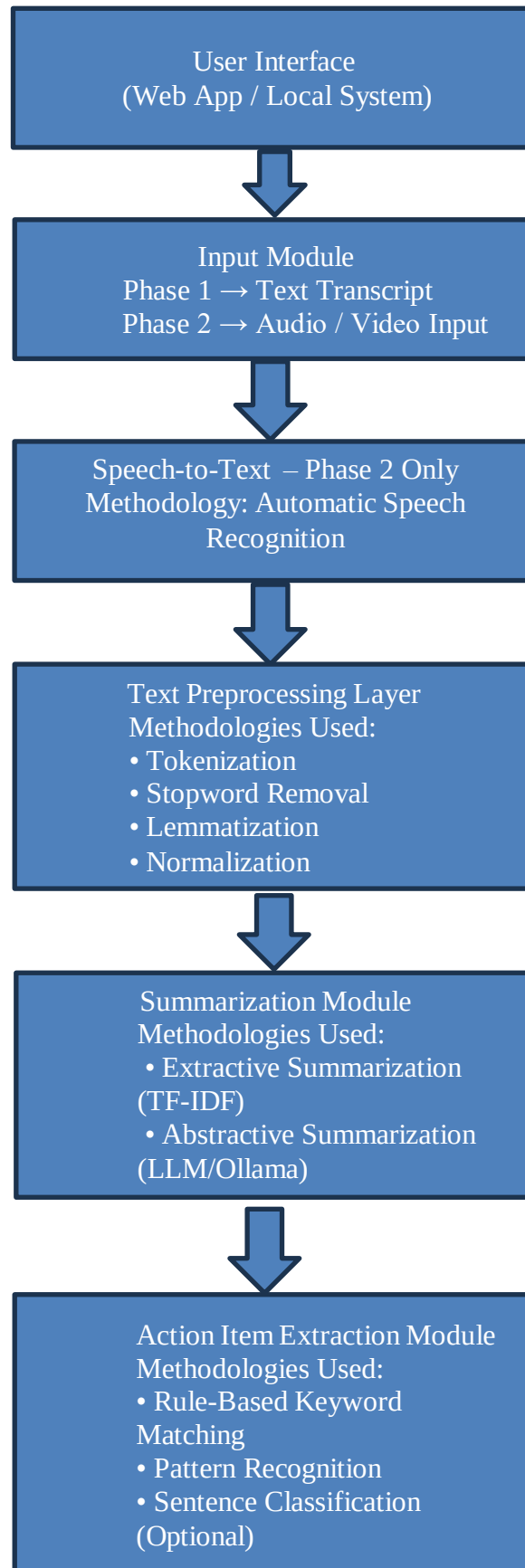
Text input in the form of :

- Meeting transcripts
 - Chat logs
 - Discussion notes
- (Audio processing is not included in Stage-I)

6. Objectives of Project

- a. To develop a system that accepts meeting text transcripts as input and automatically generates a concise summary along with identified action items.
- b. To design a module that takes meeting audio as input, converts it into text using speech-to-text processing, and then generates a summary and corresponding action items.
- c. To implement a context-aware summarization engine using a locally deployed Large Language Model (Ollama) for accurate and meaningful output generation.
- d. To store and manage generated summaries and action items in a structured format for future reference and retrieval.

7. Block Diagram of System





Output Generation

- Meeting Summary
- Action Items List
- Structured Report

8. Expected Algorithm / Processing

1. Accept meeting transcript as input
2. Perform preprocessing (tokenization, stopword removal, normalization)
3. Apply topic segmentation using cosine similarity
4. Generate multi-level summaries using TF-IDF and LLM
5. Extract action items using rule-based and NLP techniques
6. Detect decisions and risks using pattern matching + LLM
7. Perform sentiment analysis and speaker contribution analysis
8. Extract keywords using TF-IDF
9. Store results in database (SQLite)
10. Generate analytics and email report

9. Expected Output from Project

- Multi-level meeting summaries (short, detailed, executive)
- Structured action items with assignee, deadline, and priority
- Identified decisions and risks
- Speaker contribution and sentiment analysis
- Productivity score of meeting
- Extracted keywords
- Auto-generated follow-up email
- Visualization dashboard and stored meeting history

10. Abstract

In today's fast-paced professional and academic environments, meetings play a vital role in collaboration, decision-making, and project management. Organizations conduct frequent online and offline meetings to discuss plans, assign tasks, and evaluate progress. However, these meetings often generate large volumes of unstructured information, making it difficult for participants to manually record important discussion points, decisions, and action items. Traditional methods of writing meeting minutes are time-consuming, prone to human error, and often result in incomplete or inconsistent documentation.

To address these challenges, this project presents an AI-powered Meeting Summarization and Action Item Extraction System using Natural Language Processing (NLP). The primary objective of this system is to automatically convert meeting transcripts into concise summaries while identifying key tasks, responsibilities, and important decisions. By automating this process, the system significantly reduces manual effort and improves productivity.

The proposed system operates in multiple stages. In Stage-I, the focus is on processing text-based meeting transcripts. Raw meeting data is first passed through preprocessing steps such as tokenization, stop-word removal, lemmatization, and normalization to ensure clean and structured input. After preprocessing, extractive summarization techniques are applied to identify the most relevant sentences based on keyword frequency, sentence position, and semantic importance. These selected sentences are combined to generate a clear and meaningful meeting summary.

Along with summarization, the system also extracts action items using rule-based methods and pattern recognition. Sentences containing task-oriented keywords such as "assign," "complete," "submit," or "follow up" are analyzed to detect actionable information. This enables the system to automatically list tasks discussed during meetings, helping participants track responsibilities more effectively.

The project emphasizes understanding core NLP concepts including text preprocessing, feature extraction, and summary generation. A modular system architecture is designed to allow easy enhancement in future phases. The current implementation focuses on extractive summarization, but the framework supports future integration of machine learning and deep learning models for more accurate and context-aware summaries.

This system offers several advantages over traditional manual note-taking approaches. It ensures consistency in meeting documentation, minimizes information loss, and saves valuable time for professionals and students. By providing structured summaries and action lists, the system improves clarity and accountability within teams. It also supports better decision-making by allowing users to quickly review meeting outcomes.

The application of this project is wide-ranging, including corporate environments, educational institutions, online classrooms, project management teams, and remote collaboration platforms. It is especially beneficial for organizations conducting frequent meetings, where maintaining accurate records becomes challenging.

From a technical perspective, the system is implemented using Python and NLP libraries for text processing and analysis. The workflow includes data input, preprocessing, summarization, action item extraction, and output generation. The design prioritizes simplicity, scalability, and ease of integration with future components such as speech-to-text modules and task management dashboards.

Future enhancements of this project include incorporating real-time speech recognition, abstractive summarization using deep learning models, speaker identification, deadline detection, and automatic task assignment. Additional features such as meeting analytics, sentiment analysis, and visualization dashboards can further enhance the system's capabilities. These improvements will transform the application into a complete intelligent meeting assistant.

In conclusion, this project demonstrates how artificial intelligence can streamline traditional meeting workflows by automating summary generation and action item extraction. By leveraging NLP techniques, the proposed system reduces manual effort, improves documentation quality, and enhances overall productivity. The project serves as a strong foundation for building advanced AI-based meeting management solutions and highlights the growing importance of intelligent automation in modern digital environments.

✓ Sample Input 1 — Project Planning Meeting

↗ Code



Good morning everyone. Today we discussed the AI Meeting Summarizer project. Harshal explained the system architecture and overall workflow. Diksha will collect meeting transcript datasets by Wednesday. Tushar is assigned frontend development and will complete UI design by Friday. We also decided to use Python and NLP libraries for implementation. Next meeting is scheduled on Monday to review progress.

★ Extracted Action Items

Person	Task	Deadline
Diksha	Collect meeting transcript datasets	Wednesday
Tushar	Complete UI design	Friday

Literature Survey :

Sr No.	Title of Paper	Techniques Used	Key Features	Advantages	Limitations
1	Meeting Summarizer and Plan of Action Generator using NLP	TF-IDF, TextRank, PageRank, NLP	• Extractive summarization • Sentence ranking • Transcript processing	• Simple and efficient • Easy implementation • Works well on structured text	• Lacks deep understanding • No real-time processing • Limited contextual reasoning
2	AI-Driven Real-Time Summarization and Action Item Extraction in Video Conferencing Platforms	ASR, Transformer models, Deep Learning, NLP	• Real-time summarization • Action item extraction • Multispeaker handling • Integration with platforms	• High accuracy • Context-aware summarization • Suitable for real-world applications	• High computational cost • Privacy concerns • Complex implementation
3	AI-Powered Virtual Meeting Summarization System	ASR, NLP, ML, Gemini (LLM), Speaker Diarization	• Abstractive summarization • Speaker identification • Email summary sharing • Automated pipeline	• Better semantic understanding • Handles multi-speaker meetings • Improves collaboration	• Dependency on external APIs • Requires high resources • Challenges in overlapping speech

Existing systems focus mainly on summarization and action item extraction. However, they lack personalized insights for absent users, decision reasoning, confusion detection, and intelligent meeting understanding. Our proposed system aims to address these gaps by introducing advanced features such as conversation replay, decision reasoning, and user-specific meeting insights.

Drawbacks :

- Many existing systems rely heavily on extractive summarization, which only selects sentences and does not generate meaningful human-like summaries.
- Traditional approaches such as TF-IDF and TextRank fail to capture deep contextual understanding of conversations.
- Existing models struggle with multi-speaker dialogue understanding and speaker role identification.
- Long meeting transcripts cause performance degradation due to memory and token limitations in transformer-based models.
- Some systems require large annotated datasets, which are expensive and time-consuming to create.

What's New :

- Uses **LLM (Ollama)** for context-aware abstractive summarization
 - Supports **multi-level summaries** (short, detailed, executive)
 - Includes **decision detection and risk analysis modules**
 - Provides **speaker analysis and sentiment tracking**
 - Generates **productivity score and analytics dashboard**
 - Stores meeting history with **knowledge base system**
 - Generates **AI-based professional email reports**

Team Members:

Harshal Mahesh Patil (TY AIML-Roll No. 37)

Diksha Sudhakar Patil (TY AIML-Roll No. 25)

Tushar Rajendra pardeshi (TY AIML-Roll No.35)

Harshal Dyaneshwar Patil (TY AIML-Roll No.54)

